

Interesting Reference Designs Of

Title of Reference Design	Multi-Button IR Remote Control	Bluetooth Low Energy Remote Control
Image		
Key Attractions	The reference design for a multi-button IR remote control comprises both the transmitter and receiver units. A 16-pin MON08 programming interface facilitates in-circuit flash programming and debugging capabilities for seamless integration within the remote control transmitter and receiver units.	The reference design for a programmable radio-on-a-chip (PROC) BLE remote control offers versatile connectivity options, allowing it to connect to various host devices such as personal computers (PCs), smart televisions (TVs), tablets, and smartphones. The remote control design has IR-based wireless connectivity.
Highlights	The multi-button IR remote control reference design using MC68HC908LT8 microcontroller unit (MCU) from NXP Semiconductors includes transmitter and receiver units, with a significant feature being a 16-pin MON08 programming interface header. This interface allows easy in-circuit flash programming and debugging, ensuring smooth integration within the remote control units. The design meets specific performance specifications featuring low standby power consumption and varying maximum internal bus frequencies dependent on operating voltage, 4MHz at 5V and 2MHz at 3V. It houses a dual oscillator module, 8192 bytes of flash memory, and 128 bytes of RAM, tailored for air-conditioner remote controllers with functionalities like power control, temperature data transmission, and mode selection. The IR remote control transmitter, integrated into an optimised PCB layout, offers a user-friendly solution housed in a standard casing, equipped with essential features like a keypad, LCD display, and battery holder. The specially designed MCU, with dual clock drives, maintains low power usage in standby mode, essential for efficient remote control functionality. With nine buttons on the transmitter and two on the receiver, which draws power directly from the AC unit, there's no need for power-saving or standby modes on the receiver end.	The CY5672, a reference design from Cypress Semiconductor, provides a fully functional implementation of a BLE or Bluetooth smart remote control, leveraging the programmable radio-on-a-chip (PROC) BLE single-chip solution. This chip encompasses capacitive touch technology, a low-energy ARM Cortex-M0 CPU core, Bluetooth 4.1-compliant connectivity, and a built-in balun reducing the need for external parts. The versatile remote control connects easily to devices like PCs, smart TVs, tablets, and smartphones via Bluetooth Smart or USB. It features a trackpad allowing various gestures, including single or double finger taps for clicking and zooming, as well as edge swipes for scrolling. Additional features include volume and channel controls, AV input switching, and a power button, enhancing user interaction with smart TVs. Using infrared technology, the remote control offers functionalities like a power button to switch the TV on or off, volume adjustment keys, channel modification buttons for altering channels, a source key to transition between audio-video inputs, and a return key for back-navigation on a smart TV. Two AAA battery cells aligned in series power the remote control, generating a top output voltage of 3V. The entire circuitry of the remote control functions at this 2.1V voltage level.
Applications	The reference design caters to the specific application of an air-conditioner remote controller.	The reference design is suitable for various host devices such as personal computers (PCs), smart televisions (TVs), tablets, and smartphones.
OEM Brand	NXP Semiconductors	Cypress Semiconductor
URL	https://www.electronicsforu.com/electronics-projects/reference-design-for-multi-button-ir-remote-control	https://www.electronicsforu.com/electronics-projects/reference-design-for-bluetooth-low-energy-remote-control